Maternal $PM_{2.5}$ and temperature exposures pre-conception and during pregnancy are associated with increased risk of congenital Chagas disease transmission.

Matthew J. Ward PhD[1,α], Aman Patel MS[1], Shachi Mistry MPH[1], Caryn Bern MD[2], Natalie M. Bowman MD[3,4], Ayla Bullock MPH[3,4], Heather H. Burris MD[5,6], Chris Gennings PhD[1], Robert H. Gilman MD[7], Sneider Gutierrez Guarnizo PhD[7], Malasa Jois Kahn DO[8,9], Rishi Kowalski ScM[1], Priya Mallikarjuna MD[1], Sassan Noazin PhD[7], Victoria R. Rendell MD[10,11], Freddy Tinajeros PhD[7], Robert O. Wright MD[1,12,13], Nicholas B. DeFelice PhD[1,12,13], for the Chagas Disease Working Group in Peru and Bolivia[*].

Icahn School of Medicine at Mount Sinai

BACKGROUND

- Particulate air pollution (specifically <2·5μm diameter [$PM_{2\cdot5}$]), temperature, and *Trypanosoma cruzi* (*Tc*) each are deleterious to pregnancy.
- $PM_{2\cdot5}$ crosses the placenta, weakening the trophoblastic layer through inflammation.
- Sustained exposure to high heat causes chronic intrauterine inflammation, affecting placental formation.
- *Tc* parasitemia increases during pregnancy, causing placental degradation through inflammation resulting in congenital Chagas disease (cCD) in approximately one-in-twenty *Tc(+)* pregnancies.
- The interaction of $PM_{2\cdot5}$ and temperature during pregnancy with cCD is unstudied.

METHODS

- We analyzed a pregnancy cohort (2006-2014) including 581 *Tc(+)* pregnancies with 46 congenitally infected infants from Santa Cruz & Camiri, Bolivia.
- European Space Agency-Copernicus estimates of monthly mean wet-bulb temperature and $PM_{2\cdot5}$ were used to create individual monthly pregnancy exposures for 12-months prior-to-birth.
- Lagged-Weighted Quantile Sum (LWQS) regression analysis was conducted to determine the time-varying association of these exposures with cCD.
- 'Months Since Birth' with units of -12 to 0 months; 0 indicates birth with a bootstrap ensemble step.
- A time-varying WQS(t) index calculated for each infant using estimated weights. Final weights for the two mixture components were the average over the time-specific 100 bootstrap samples.
- A reverse distributed lagged regression model analysis was conducted on the association between the binominal outcome variable (+/- for *Tc*) and the estimated WQS(t) indices, adjusted for age of mother and sex of baby.

DATA

Table 1. Data sources for environment and LHRF in Santa Cruz, Bolivia.

Variable	Platform	Source	Spatial Resolution	Temporal Resolution
Temp: Wet Bulb, Surface	ERA5-Land[35]	ESA - Copernicus C3S	0.1 °	Hourly, 2005 - present
$PM_{2.5}$				
LHRF Data[*]	Cohort	Questionnaire*	Dyad - Neighborhood	Enrollment, follow-up visits

*Includes data on neighborhood, household size, type, and occupancy, household education and occupations.

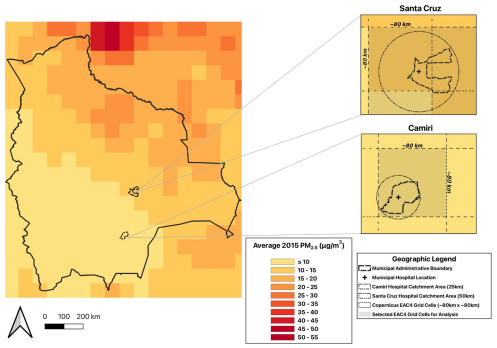

Figure 1. Left: Bolivia. Red Gradient: Mean monthly PM2·5 levels for 2015. Upper Inset: Santa Cruz municipality (dashed polygon), Hospital Universitario Japones (black cross), Copernicus grid cells averaged for exposure values (4 dotted squares), and 50 km catchment radius (dotted circle). Lower Inset: Camiri municipality (dashed polygon), Municipal Hospital of Camiri (black cross), Copernicus grid cell for exposure values (1 dotted square), and 50 km catchment radius (dotted circle).

RESULTS

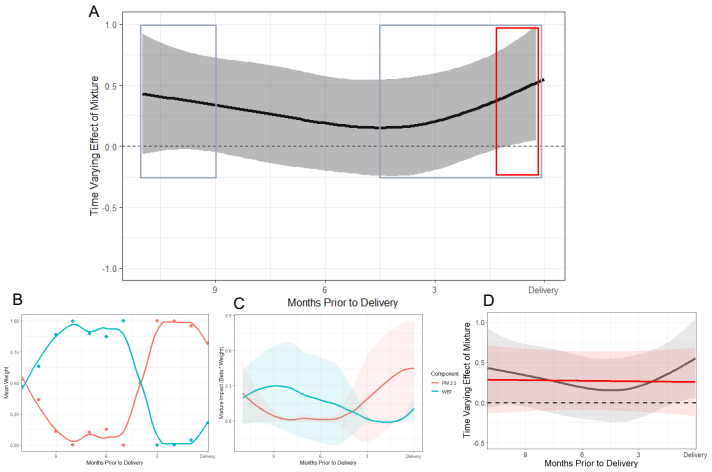

Figure 2. Panel A: Black line & grey CI: Significant association parameter with corresponding 95% CI. Blue boxes: Positive trends associated with increased cCD. Red box; significant critical window. **Panel B:** Estimated weights of each parameter at each time point in the L-WQS model (blue = WBT, red = $PM_{2\cdot5}$). **Panel C:** The weighted association (estimated weights multiplied by the association parameter at each time point) for both $PM_{2\cdot5}$ (red) and temperature (blue). **Panel D:** Black line & grey CI: Significant association parameter with corresponding 95% CI. Red line & red CI: Null model with perturbed environmental exposures.

RESULTS

- The weighted association for both $PM_{2.5}$ and temperature indicated a positive association pre-conception and prenatally.
- Significant positive trend begins near -5 months with a **critical window between -2 months and birth**.
- Positive trend (blue box: left) beginning at -12 mo until end of 1st trimester with a nearly significant window at 12 till conception.
- Temperature contributed to ~25% of the weighted associations and $PM_{2.5}$ ~75%.
- cCD was significantly associated with exposure to higher temperatures and $PM_{2\cdot5}$ pre-conception and high $PM_{2\cdot5}$ in the last three months of pregnancy.

Interpretation Systemic inflammation from $PM_{2\cdot5}$ and high temperatures likely depress the already modulated maternal immune system allowing higher parasitemia during pregnancy. Inflammatory responses to $PM_{2\cdot5}$, temperature, and *Tc* may weaken the placental barrier, allowing the parasite to more easily cross the marginal sinus and the trophoblastic layer of the intervillous space and infect the fetus. In urban areas, vector-bone *Tc* transmission is now uncommon; however, there is still a significant incidence of cCD. Mitigating the effects exposure to high $PM_{2\cdot5}$ and temperatures during pregnancy is an important component to preventing cCD.

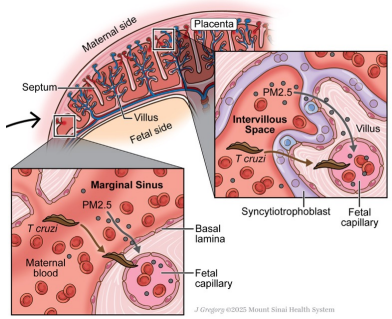

Figure 3. Illustration of parasite invasion of the fetal blood supply by crossing the basal lamina of the marginal sinus and trophoblast layer of the intervillous space.

FUNDING & COLLABORATORS

[1]Department of Environmental Medicine, Icahn School of Medicine at Mount Sinai, New York, New York, USA. [2]Department of Epidemiology and Biostatistics, University of California, San Francisco, CA, USA. [3]Division of Infectious Diseases, University of North Carolina at Chapel Hill, Chapel Hill, NC, USA. [4]Institute of Global Health and Infectious Diseases, University of North Carolina at Chapel Hill, Chapel Hill, NC, USA. [5]Department of Pediatrics, Perelman School of Medicine, University of Pennsylvania, Philadelphia, Pennsylvania, USA. [6]Division of Neonatology, Children's Hospital of Philadelphia, University of Pennsylvania, Philadelphia, Pennsylvania, USA. [7]Department of International Health, Johns Hopkins Bloomberg School of Public Health, Baltimore, Maryland, USA. [8]Department of Family Medicine, Brown University, Providence, Rhode Island, USA. [9]Kahn Health PC, Montclair, New Jersey, USA. [10]Duke Global Health Institute, Duke University, Durham, North Carolina, USA. [11]University of Wisconsin School of Medicine and Public Health, Madison, Wisconsin. [12]Institute for Exposomic Research, Icahn School of Medicine at Mount Sinai, New York, New York, USA. [13]Arnhold Institute for Global Health, Icahn School of Medicine at Mount Sinai, New York, New York, USA. NIH: T32HD049311, P30ES023515, K25HD109509, R21AI072093, R01AI087776, R01AI151295, R24TW007988, D43TW009340, T35AI065385, D43TW006581. [α]Corresponding Author: Matthew J. Ward, PhD; matthew.ward@mssm.edu